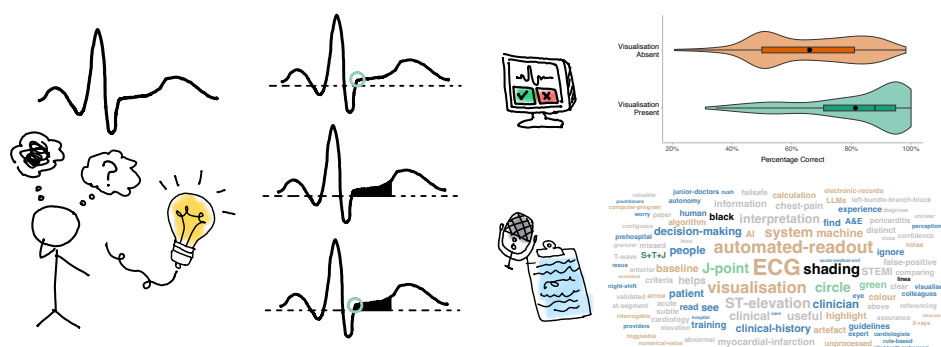


Graphical Abstract

Increasing the accuracy of ST-segment elevation detection in an ECG through automated visual rules

Lukas Hughes-Noehrer, Gabriel Strain, Adina Rahim, James Sinnott, Jonathan Carlton, Crystal T. Wu, Richard Body, Caroline Jay



Increasing the accuracy of ST-segment elevation detection in an ECG through automated visual rules

Lukas Hughes-Noehrer^{a,*}, Gabriel Strain^a, Adina Rahim^b, James Sinnott^b, Jonathan Carlton^a, Crystal T. Wu^a, Richard Body^c, Caroline Jay^a

^a*The University of Manchester, Department of Computer Science, Oxford Road, Manchester, United Kingdom, M13 9PL*

^b*The University of Manchester, Research IT, IT Services, Oxford Road, Manchester, United Kingdom, M13 9PL*

^c*The University of Manchester, Division of Cardiovascular Sciences, Oxford Road, Manchester, United Kingdom, M13 9PL*

Abstract

Objective: Accurate identification of ST-segment elevation on the electrocardiogram (ECG) is essential for diagnosing acute myocardial infarction, yet interpretation remains challenging due to noise, signal variability, differences in user expertise and limitations of existing automated methods. This study investigates whether novel, clinically motivated visualisation techniques – including a novel heart-rate-corrected ST-analysis window (STc) – can make ST-segment elevation easier to detect.

Methods: We conducted a mixed-methods investigation comprising two components: (Part I) qualitative validation interviews with practising clinicians from emergency medicine, anaesthesia, critical care and cardiology, followed by (Part II) a preregistered online experiment with members of the public unfamiliar with ECG interpretation.

Results: In the clinician interviews ($n = 6$), participants described the visualisations as transparent, clinically coherent, helpful for training, and complementary to automated prompts. Clinicians particularly valued visual

*Corresponding author

Email address: `lukas.noehrer@manchester.ac.uk` (Lukas Hughes-Noehrer)

cues that provided a consistent and physiologically grounded ST-analysis window; for this reason, we included STc, a heart-rate-corrected interval that avoids reliance on arbitrary fixed millisecond cut-offs and mitigates heart-rate-driven distortions in defining the end of the ST-segment.

In the online experiment ($n = 303$), participants classified single-lead, single-beat ECG images that either incorporated visualisations or displayed the visually unannotated signal. Accuracy and response times, analysed using generalised linear mixed models, showed that participants made significantly more accurate and faster decisions when visualisations were present. %Performance differed across visualisation types: shading-based designs were most effective when ST-segment elevation was present, while J-point markers were particularly helpful when elevation was absent.

Conclusion: The findings demonstrate that intuitive visual cues improve detection of ST-segment elevation, and could therefore provide a form of novel, explainable, automated analysis for supporting future clinical decision support tools, training resources, and patient-facing applications.

Keywords: electrocardiogram, ST elevation, visualisation, crowdsourced, user validation

1. Introduction

Cardiovascular diseases remain the leading cause of death worldwide, with myocardial infarction (MI), commonly known as heart attack, representing a major health concern [1, 2]. The electrocardiogram (ECG/EKG) is an important diagnostic tool, which provides visualisations of the heart’s electrical impulses. ECGs are quick, cost-effective, and non-invasive, and are therefore usually the first investigation a patient presenting with a suspected cardiac issue undergoes [3].

Despite their ubiquity as a diagnostic tool, ECGs are difficult to interpret [4, 5, 6]. ECG interpretation accuracy varies, even amongst those trained in cardiology. A 2020 meta-analysis of 78 studies across all levels of clinical training reported a median accuracy rate of 54% [5]. As medical professionals gain experience and additional training, accuracy trends upwards. Yet, resident doctors (those still in training) make up significant proportions of medical workforces [7], and errors in ECG interpretation account for poor

health outcomes and preventable deaths [8]. Improving ECG interpretation remains a critical issue.

Computer-assisted interpretation of ECGs dates back to the 1950s [9], and algorithmic support is now standard in ECG monitoring systems. Contemporary Computerised Interpretation of ECG (CIE) aims to reduce diagnostic errors and support clinical decision-making, yet studies have demonstrated its fallibility; clinician oversight is required to avoid misdiagnosis [10, 11]. Advances in Artificial Intelligence (AI) and Machine Learning (ML), coupled with large waveform datasets, have enabled methods ranging from rule-based systems to deep learning approaches [12, 13, 14]. Despite promising clinical trials and regulatory approval of some devices [15], adoption remains limited due to systemic bias, liability concerns, cybersecurity risks, and unresolved technical and ethical challenges [16, 17, 18, 19, 20]. Trust in AI systems remains a critical barrier, with clinicians expressing cautious optimism, but highlighting issues around cost, usability, and explainability [21, 22, 23, 24]. Automated techniques that can reliably support the intuitive understanding of ECG data are therefore particularly important.

Building on our recent work demonstrating that explainable visualisation techniques are useful for facilitating the interpretation of ECGs and aiding clinical decision-making [25, 26], we developed novel ST-segment elevation (ST-elevation) visualisations informed by routinely used clinical rules [27]. Our visualisations highlight clinical areas of interest to support efficient, accurate ECG interpretation. To assess our visualisations, we adopted a mixed-methods approach: a clinical user validation – presented in Part I – and a reproducible single-experiment study with lay participants – presented in Part II. We selected lay people to evaluate the usefulness of our visualisation under the assumption that if individuals with no prior experience in interpreting ECGs can accurately identify ST-elevation, it provides strong evidence for the comprehensibility and the intuitiveness of our explainable designs.

1.1. Hypotheses

Our Part II study tested three visualisation types with the aim of scaffolding understanding and interpretation of ST-elevations. Owing to the novelty of the designs and the testing of lay populations with ECG visualisations, we do not hypothesise that one design may be superior to another. Previous evidence that novel designs can be beneficial with regards to the interpretation

of ECG data among lay participants [28, 29, 30, 31, 32] lead to the following hypotheses:

- H1: Participants’ performance on a binary decision task recognising the presence or absence of ST-elevation will be more accurate when a visualisation condition is present as opposed to when there is no additional visualisation beyond what is standard for ECG data.
- H2: Participants will make binary classification decisions significantly more quickly when additional visualisations are present compared to when they are not.

Table 1: Statement of Significance.

Problem or Issue	Accurate identification of ST-segment elevation is critical for diagnosing acute myocardial infarction, yet ECG interpretation remains difficult due to variability, noise, and inconsistent clinician agreement.
What is Already Known	Both clinicians and automated systems can struggle with ECG interpretation; existing AI tools often lack transparency, limiting trust and adoption despite advances in computational methods.
What this Paper Adds	We develop explainable, rule-based ECG visualisations, including a heart-rate corrected ST analysis window (STc), and show through clinician validation and a pre-registered online experiment that these visual cues significantly improve detection accuracy and speed.
Who would benefit from the new knowledge in this paper	Clinicians, trainees, and developers of clinical decision-support systems benefit from improved interpretability, training potential, and scalable design principles for trustworthy, explainable ECG analysis.

2. Related Work

2.1. ST-Elevation Myocardial Infarction

MI refers to a partial or complete blockage of the blood flow to a portion of the heart muscle [33]. This blockage deprives parts of the heart of oxygen, %potentially leading to cell death andnecrosis. and depending on the extent,

may appear asymptomatic or progress to sudden death. Types of MI are diagnosed based on the presence of cardiac biomarkers in the blood concurrent with other criteria, including changes in the electrical activity of the heart [34] detected via ECG. In a clinical setting, 10 electrodes are attached to different parts of the body to produce 12 different “views” of the heart’s electrical activity [3] – referred to as *leads*.

ST-segment Elevation Myocardial Infarction (STEMI) is diagnostically defined as [27, 35]:

The presence of an ST-segment that is elevated above the baseline at the J-point by at least 0.1 millivolts (mV) in any two contiguous leads except V2 and V3. In leads V2 and V3, this elevation must be at least 0.2 mV in males aged 40 years or older, at least 0.25 mV in males aged under 40 years, and at least 0.15 mV in females.

Here, *contiguous* refers to leads that represent adjacent anatomical areas. Heart attacks featuring STEMI are differentiated from those where an ST-elevation is absent (NSTEMI); the former is more severe, and so is studied in this work. Survival following a STEMI has improved substantially over the past few decades [36, 37, 38]. Despite this, STEMI and other forms of MI remain a leading cause of mortality [1, 2] and lower Health-Related Quality of Life (HRQoL) [39, 40, 41]. Prompt diagnosis and treatment are crucial for reducing mortality, limiting hospital stay duration, and for prognosis following a STEMI event [2].

2.2. ECG Presentation and Existing Visualisation

The way in which ECG data are presented to clinicians has changed little since its invention over a century ago [42]. The quintessential form of the ECG and the issues that clinicians face when engaging with it remain much the same. ECGs are typically drawn on a Cartesian line graph with the signal voltage in millivolts on the (y)-axis and time in milliseconds (ms) on the (x)-axis. A background grid system aids interpretation; Figure 1 illustrates this system and a typical ECG signal featuring the P-wave, the QRS complex, the ST-segment, the T-wave, and the J-point. Whether produced manually by a pen on grid paper or drawn digitally by a computer, the resulting

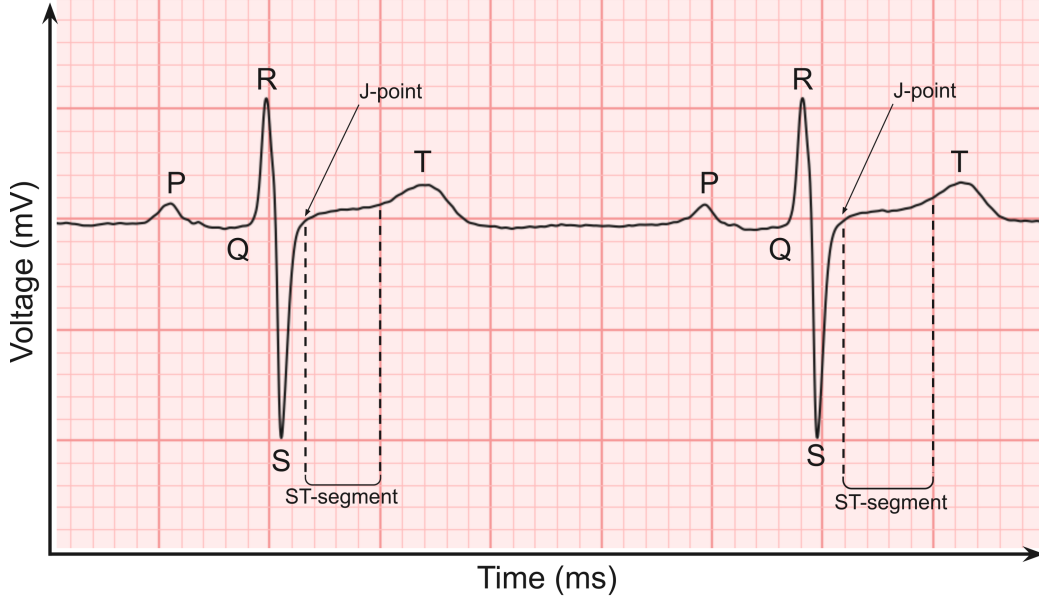


Figure 1: Illustration of an ECG including the P-wave, the QRS complex, the ST-segment, the T-wave, and the J-point. Each small red square represents 40 milliseconds (ms) or 0.1 millivolts (mV). This signal was taken from the V4 chest lead.

visualisation is essentially identical. Clinicians are typically not assisted by any visual aids such as labels or markings when analysing ECGs, besides the name of the leads, and if available, computer-automated interpretations presented as text.

Chest and limb leads have been combined into separate 3D representations of cardiac electrical activity [43, 44]. This approach has gained some clinical acceptance [43] and has been used in algorithmic classification [44, 45, 46]. Studies of “inverted” ECG leads show mixed results; however, Perron et al. [47] found that adding seven inverted leads improved detection of induced arterial blockage. Other approaches include glyph-based interactive systems [48], vectorial methods representing magnitude and direction of cardiac electrical signals [49, 50, 51, 52], and spatial mapping of body-surface electrical potentials [53, 54]. While useful, these techniques are generally considered supplementary to standard ECG visualisations [55], as they increase data volume or complexity and are less suitable for non-experts. In contrast, our approach focuses on enhancing the salience of relevant features rather than adding information.

2.3. Using Clinical Rules to Guide Visualisation Design

Clinical decision rules and protocols guide healthcare data visualisation by translating clinical knowledge into structured representations. These rules often take the form of thresholds, scores, or pathways that support detection of abnormal states, care escalation, and coordinated response, informing clinical decisions [56]. In medical informatics, effective visualisations help overcome information overload and enable recognition of clinically meaningful changes compared to raw data alone [57]. Visual aggregation and structured timelines further improve consistency in interpreting complex data [58] and enhance situational awareness (SA) in clinical practice [59]. Effectiveness depends on how information is visually encoded. Design choices shape perception and reasoning; poor encoding can undermine decision-making even when underlying data are sound [60]. Design studies show that mismatches between visual representations, user tasks, and context often lead to poor adoption or misuse [61].

Visualisations are also used to communicate health data to lay audiences [62]. Here, decision rules are often embedded in summaries, indicators, or feedback, shaping both perception and action. This highlights the need to align design with audience needs and capabilities. Clear communication goals, taxonomy-guided design, and audience-specific evaluation are essential for successful visualisation [63]. The effectiveness of clinical decision rules therefore depends not only on their logic but also on their visual representation, which determines understanding, trust, and appropriate action.

3. Methods

This section outlines our research methodology, encompassing visualisation design, the collection of clinician feedback on the visualisations, and the lay-person experiment. Details of algorithm development is presented in Appendix A.

3.1. Visualisation Design

The visualisations were designed to increase the salience of the visual features relevant to ST-segment elevation analysis and the experimental subtasks outlined in Section 3.4. To reduce the cognitive load of ECG interpretation

tasks, we applied strategies taken from Cognitive Load Theory (CLT) [64]. After iterative cycles of discussion regarding visualising clinical rules for ST-elevation during the Responsible Research and Innovation (RRI) workshops, a final set of three visualisation types was decided. Examples are shown in Figure 2.

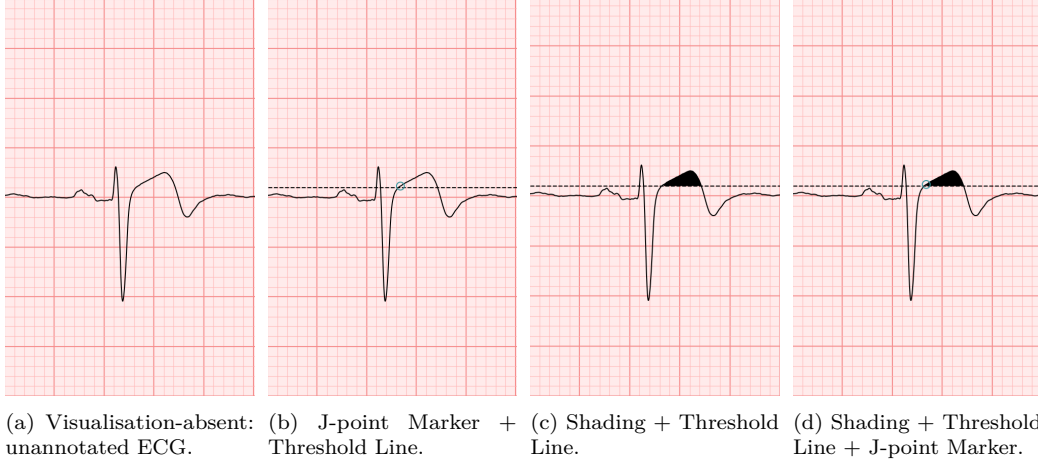


Figure 2: Stimuli Examples: examples of the stimuli used in the experiment in visualisation-absent and the three visualisation-present conditions. This ECG signal is taken from the V4 chest lead and has an elevated ST-segment.

All visualisation types feature a dashed threshold line at 0.1 mV above the baseline. The inclusion of this line was due to reports that very low levels of ST-elevation may be responsible for disagreements between clinicians when assessing ST-elevation on an ECG [65]. Threshold-line-only or threshold-line + baseline designs were ruled out during the RRI workshops as they caused uncertainty in interpretation. In addition to the threshold line, we took two approaches to address another source of potential disagreement: the location of the J-point [66]. The first, *J-point Marker + Threshold Line*, employs a green marking circle highlighting the J-point on the ECG signal. The colour was chosen to be visible to those with colour-blindness against the red grid background. The second, *Shading + Threshold Line*, removes viewers' obligation to find a J-point, instead using black coloured shading to display the proportion of the ST-segment sitting above the threshold line. The final visualisation type, *Shading + Threshold Line + J-point Marker*, combined these approaches. Markers for the threshold line, the J-point, and the shading can be applied on top of the signal individually, as an ensemble, or omitted,

to obtain ECG images with and without visualisation.

3.2. Novel Correction of ST-segment Calculations

There is long-standing disagreement when defining the J-point and varying practice in defining the ST-segment to detect ST-segment elevation [67]. Fixed time cut-offs, such as 40 ms, 60 ms, or 80 ms after the J-point, have been widely used, yet their physiological validity is contested and diagnostic performance is sensitive to chosen offsets [68]. As ventricular repolarisation shortens at high heart rates and lengthens at low heart rates through rate-dependent action potential adaptation, a fixed delay from the J-point does not consistently capture the same repolarisation phase across recordings. As a result, a fixed cut-off to define the diagnostically-important part of the ST-segment may introduce heart-rate-dependent distortions in both visual and algorithmic analysis.

Given these limitations and driven by the need to define a diagnostic window of the ST-segment for the shading-based visualisation introduced in Section 3.1, we developed a novel algorithm to calculate the ST-segment, STc (ST corrected), inspired by Bazett’s formula to correct calculation of the QT interval [69], which is calculated as $STc = ST/\sqrt{RR}$, where STc is in milliseconds, ST is the duration between the J-point and the T-wave onset (ST-segment) in milliseconds, and the average R-peak to R-peak interval RR in seconds.

As STc uses a Bazett-type correction, 60 bpm serves as the reference point ($RR = 1$ s) and removes the heart-rate-dependent distortion. $\%STc$ therefore creates a heart-rate-normalised window that aligns with the same repolarisation phase at any heart rate, without altering the ECG signal or imposing a specific physiological state. The resulting STc provides a cycle-length-normalised, physiologically coherent window that is comparable across beats, patients, and varying heart rates. Unlike fixed cut-offs, which risk sampling different electrophysiological phases depending on heart rates, STc ensures that the analysis window stays consistent. In practice, STc defines the duration of the ST-segment, determining its endpoint relative to the J-point. The resulting heart-rate-normalised diagnostic window is overlaid in the time domain as a visualisation on the ECG signal, whose temporal structure remains heart-rate-dependent.

By combining morphology-based delineation with heart-rate correction, *STc* provides an alternative to conventional post-J-point offsets and may yield a more reliable and interpretable framework for visualising and analysing ST-segment behaviour in clinical and computational applications.

3.3. Part I: Clinical Validation of Visualisations

We conducted a qualitative evaluation of the usefulness and relevance of our visualisation designs. Participants across four specialities (Anaesthesia, Critical Care, Emergency Medicine, and Cardiology) were convenience sampled [70] as part of their ongoing stakeholder engagement in the project. These specialities were selected as they regularly interpret ECGs. All clinicians were based in the UK at the time of evaluation, and Appendix B outlines the distribution of participants by speciality and training level. We aimed to recruit at least five clinicians, a common threshold for formative medical device and software evaluation [71], and our sample size followed established human-factors principles where small expert samples are appropriate for formative, low risk evaluations [72].

Based on the Ethics Decision Tool at the University of Manchester and the NHS Health Research Authority Ethics Toolkit [73], the study was exempt from ethical review as we solely consulted with professionals (i) strictly within their professional remit, and (ii) did not collect any personally identifiable information apart from those on the exempt list, i.e. names, professional roles, signed consent, and audio recordings.

The evaluations were undertaken between November 2025 and January 2026 via the video conferencing software Microsoft Teams [74] (video function disabled). Participants were included on the basis that they are currently employed clinicians with “experiential relevance” [75, p.124] in terms of ECG usage and clinical knowledge. As we aimed to generate data that establishes a representational account of participants’ professional roles, we refrain from including the names of clinicians. All participants received a Participant Information Sheet (PIS) prior to the interview and provided informed consent. Participants were further given the option of withdrawing data up to 14 days after the date of participation, and received a shopping voucher worth Fifteen Pounds Sterling (GBP) as a token of appreciation.

The evaluations followed a semi-structured guide [76] based on generative questions [77] to encourage replies in an open format. The guide was developed

during RRI workshops and is grounded in practice-based approaches of Human-Computer Interaction (HCI) to enable investigation of participants’ lived experiences with technologies [78]. The guide introduced our visualisations, explained their design, and detailed their explainability based on clinical rules. We then asked about their clinical relevance, usefulness, and clinicians’ preferences, followed by considerations of how they would use them in clinical practice.

3.4. Part II: Online Experiment with Lay Participants

Our goal was to design visualisations based on clinical rules to enable intuitive visual identification of ST-elevation – as commonly prevalent in STEMI – on ECGs. Motivated by previous work [32], we conducted rounds of RRI workshops with lay people, clinicians regularly interpreting ECGs, and HCI experts to draft the research pathway.

Since we recruited lay participants unfamiliar with ECGs, asking participants to spot a STEMI on an ECG, with or without the added visualisation, would be a task with a high intrinsic cognitive load. Instead, we segmented this task into smaller, more easily achievable steps:

1. Solely focused on ST-elevation without introducing the concept of STEMI.
2. Removed the multiple lead contiguity rule.
3. Only showed single-heartbeat single-lead images.
4. Only showed leads where the 0.1 mV rule applies.

Participants were trained on these principles, and were not introduced to concepts or rules unnecessary for the completion of the task. The task can be broken down into two subtasks: 1) locate the J-point, and 2) check for the ST-elevation, *in purely descriptive terms based on the ECG images shown*.

3.4.1. Modelling

Our primary outcome measure – participants’ performance on the ST-elevation classification task – is binary. We built Generalised Linear Mixed Models (GLMMs) of the binomial family for analysis. All models featured random intercepts for items and participants, and where appropriate, random slopes for participants or items were specified with regard to the effects of

predictors. The **buildmer** package (version 2.12 [79]) was used in R (version 4.5.1 [80]) to automate model syntax selection using stepwise regression. On provision of a maximal model, **buildmer** identifies the most complex model that successfully converges. The **lme4** package (version 1.1-37 [81]) supplies the **glmer()** function to **buildmer** that constructs the GLMM.

3.4.2. Stimuli

Our stimuli were designed based on the visualisation pipeline outlined in Appendix A and the core principles of our experiment described in Section 3.4. The stimuli used in the experiment consisted of static ECG tracings representing single heartbeats from any of the included leads (I, II, III, aVR, aVL, aVF, V1, V4, V5, and V6), displayed either with or without visualisations (see Figure 2). ECG segments were sampled to illustrate the presence or absence of ST-elevation.

Several assumptions underlie the stimuli design. First, ECGs were assumed to have sufficient signal quality for identifiable landmarks (baseline, J-point, ST-segment), ensured via manual curation and clinical verification. Second, single-lead, single-heartbeat images removed temporal and multi-lead context to isolate perceptual interpretation rather than to facilitate clinical diagnosis. Third, static images were treated as adequate proxies for visual interpretability, despite real-world practice relying on longer, multi-lead data. Finally, visual parameters (e.g., scaling, grid, thresholds, image size) were held constant so differences reflect visual cues, not presentation.

These assumptions reflect a controlled design to test visualisation for ST-elevation interpretation and enable systematic comparison, rather than replicate real-world complexity.

3.4.3. Procedure

The experiments were built using PsychoPy [82] and hosted on Pavlovia.org. Participation was permitted using laptop or desktop computers only. A flowchart depicting the experimental procedure is presented in Figure 3. The experiment received ethical approval from the Proportionate University Research Ethics Committee at the University of Manchester with reference 2024-21655-38287 and study conduct adhered to the Declaration of Helsinki. After viewing the PIS and providing consent, participants provided their age and gender identity. A 4-minute training module following, outlining the

general form of ECG, and ST-elevation rules within the experimental context (see Section 2.1 and Section 3.4). The final slide of each training module instructed participants on how to use the visualisation they were assigned to (see supplemental materials). Four practice trials were provided which provided participants feedback on *why* they were correct or incorrect.

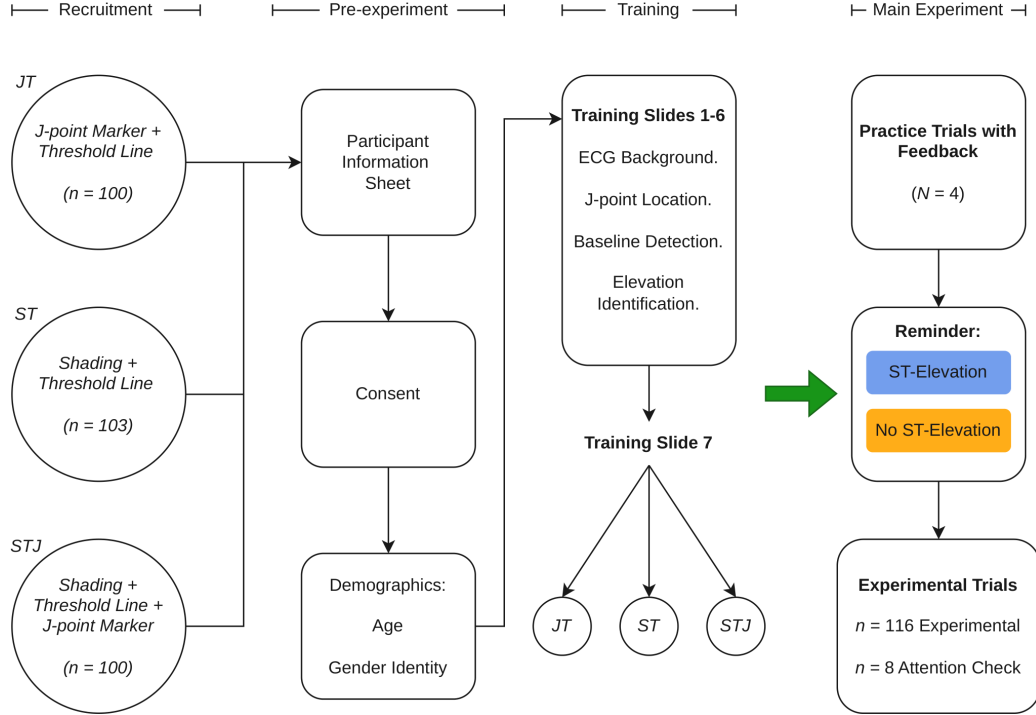


Figure 3: Experimental flowchart depicting the stages of the experimental procedure.

Participants were reminded to press the “ST-Elevation” button or “No ST-Elevation” button if they believed the ECG did or did not show ST-elevation. Participants were warned that the buttons’ positions would change throughout the experiment. Participants then worked through a series of 124 trials; an example is provided in A.2. Each was preceded by text that read “Please look at the following ECG and decide if ST-elevation is present.” ($n = 116$, black, experimental trials) or “Please IGNORE the following ECG and click on the centre of the visualisation.” ($n = 8$, red, attention check trials). There was no time limit per trial. Participants who successfully completed the experiment were paid pro rata through Prolific at a rate of 6 GBP per hour.

All participants were recruited with the Prolific platform [83]. We sought

to recruit $n = 100$ participants for each of the three between-participants experiments. English fluency, normal or corrected-to-normal vision, and having not taken part in any of the RRI workshops described in Section 3.4 were required for participation. Participants were also instructed not to take part if they had prior medical knowledge or experience interpreting ECGs.

3.4.4. Experimental Design

We employed a mixed design. Within participants, all saw the same 58 ECG tracings without visualisations (29 with ST-elevation, 29 without). The between-participants factor was the type of ECG visualisation utilised, for the same 58 ECG tracings.

4. Results

4.1. Clinician Preference for Distinct Visual Cues

All clinicians consistently identified the green circle marking the J-point as the most effective and reliable cue for detecting ST-elevation. Its distinct colour and constant presence across conditions made it easy to detect, even in noisy or artefact-prone ECGs. Its reliability – through continuous display – reassured users that the algorithm was functioning correctly. The threshold (dashed) line was also valued as a clear, guideline-based reference.

Shading provided an immediate visual cue, making elevation obvious. However, some clinicians felt it could obscure waveform details, with one noting that “the black actually takes away from highlighting” (P2), and another describing the combination with the green circle as “the best and the worst of both” (P4), balancing clarity with potential clutter. Others emphasised over-sensitivity as beneficial: P6 argued visualisations should favour false positives, especially in high-pressure or low-experience settings, viewing prominent cues as safety features. P6 also highlighted the combined visualisation as most informative, reducing cognitive effort while signalling abnormal morphology. Despite this, many clinicians suggested shading should be optional or toggleable to suit user preference.

4.2. Clinician Trust in Guideline-Based Visuals

There was strong consensus that visualisations grounded in established clinical guidelines, such as those from the American Heart Association (AHA) or European Society of Cardiology (ESC), fostered greater trust and transparency. Clinicians appreciated that these rule-based tools could be easily referenced and updated in line with evolving standards. P1 stated that “international guidelines always back you up”, suggesting confidence in the system’s recommendations. The ability to trace the logic of the visualisations back to a specific guideline was seen as particularly valuable for maintaining clinical assurance and regulatory compliance (P4).

4.2.1. Role in Contemporary Computerised Interpretation of ECG

Our visualisations were widely regarded as a valuable complement to Contemporary Computerised Interpretation of ECG (CIE). Clinicians reported that visualisations helped clarify algorithmic outputs, especially when the automated printout was ambiguous or difficult to interpret. One participant explained, “having both the interpretation and the visualisation is really useful [...] it lets you see what it’s [the CIE] talking about – or what you missed” (P3). Visual cues were also seen as helpful for confirming or questioning the automated diagnosis, supporting more confident and informed decision-making (P4). One clinician highlighted that CIE often adds bias by suggesting a possible diagnosis rather than simply highlighting a feature, and added that a visualisation “prompts a clinician in an unbiased way [...], because it doesn’t take away the human’s autonomy. It doesn’t take away the human decision-making factor” (P2).

4.2.2. Educational Value

The educational benefits of visual cues were repeatedly highlighted. Clinicians described how these tools could support less experienced colleagues and trainees, helping them to develop ECG interpretation skills and clinical judgement. P2 remarked that “prompting with cues actually will help in terms of training, learning, and retaining clinical autonomy”. The ability to toggle visualisations on and off was seen as particularly useful for educational purposes, allowing users to test their own interpretation before revealing the algorithm’s guidance (P4).

4.2.3. Universal Application and Pre-Hospital Utility

There was a clear preference for universal application of these visual aids across all ECGs, rather than restricting them to borderline cases. Clinicians felt that routine use would help practitioners become familiar with the cues, standardise interpretation, and reduce variability in practice. This standardisation was also seen as enhancing patient safety by lowering the risk of missed or misclassified ST-segment elevation and prompting timely escalation, particularly for less experienced staff (P3). Importantly, several clinicians noted that J-point localisation can be difficult and “subjective” (P1), with further remarks on inter-observer variability: “ask 20 consultants, you get 10 different J-points” (P3) and baseline artefacts complicating identification (P2). The potential utility for pre-hospital providers was also highlighted, with one participant describing how a clear, rules-based visualisation could support rapid decision-making and streamline care pathways: “this visualisation would be a very clear rules-based ‘Yes, this meets the criteria!’” (P4).

4.3. Lay Participants Performance on the Binary Classification Task

105 participants completed the *J-point Marker + Threshold Line* experiment; 5 participants failed more than 3 out of 8 attention check questions and therefore had their submissions rejected in accordance with pre-registration stipulations, yielding a final sample of 100 included in the analysis. 107 participants completed the *Shading + Threshold Line* experiment; 4 were rejected, resulting in 103 samples. 111 participants completed the *Shading + Threshold Line + J-point Marker* experiment; 11 were rejected, resulting in 100 samples. 152 participants self-identified as male and 151 as female. Participants’ mean age was 37.5 ($SD = 13.2$) years.

Figure 4 contains a summary of the results across the whole experiment. Each experimental condition is plotted separately, with standard errors plotted as error bars. To investigate the first hypothesis, that participants’ performance on the ST-elevation classification task would be more accurate when a visualisation is present, we built a model that uses the presence or absence of visualisation to predict performance. This model featured random intercepts for items and participants, and random slopes for participants with regard to the presence or absence of a visualisation. A likelihood ratio test revealed that the model including the presence or absence of a visualisation explained significantly more variance than a null model ($\chi^2(1) = 77.37$, $p < .001$).

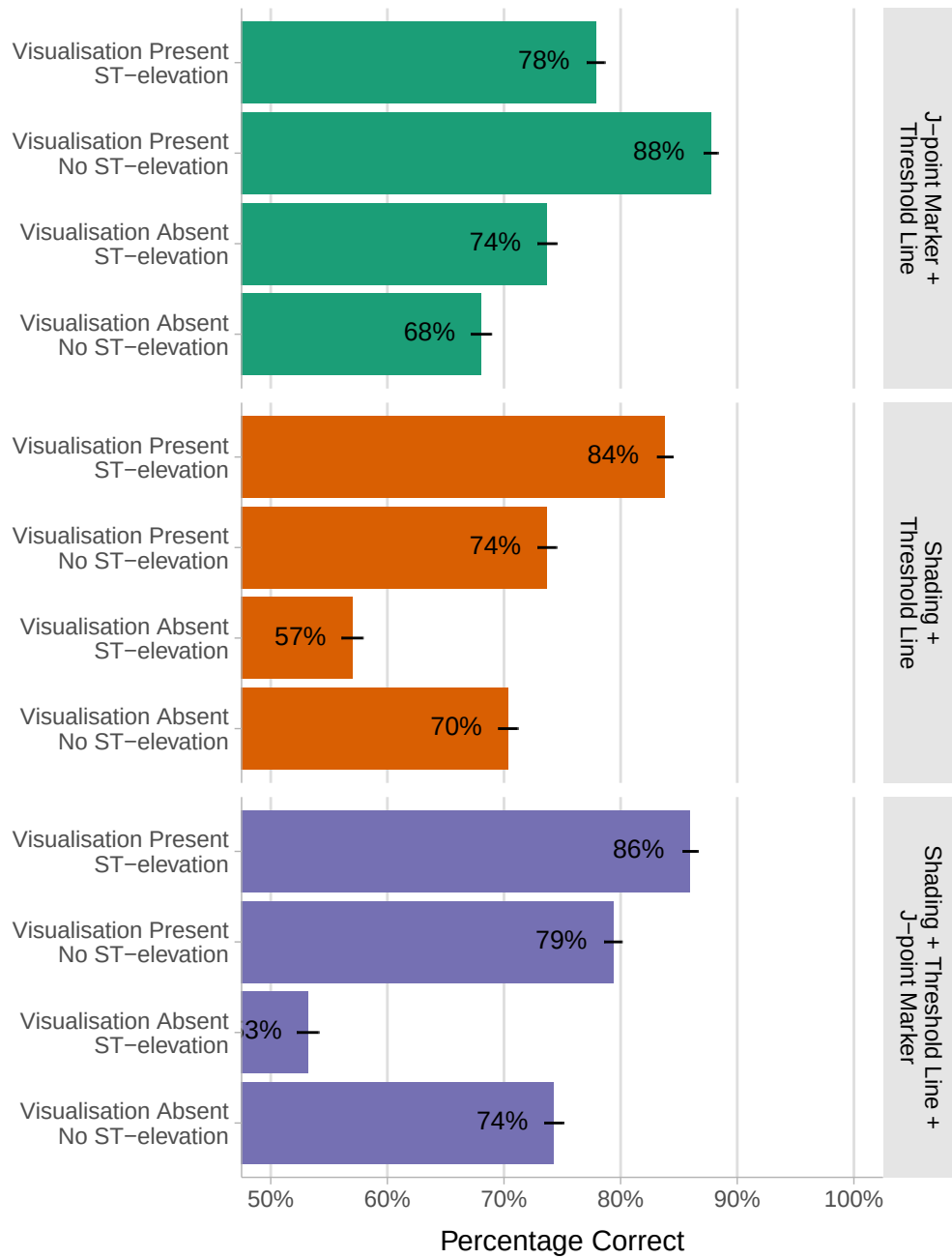


Figure 4: Mean Percentage Correctness Statistics: bars show mean percentage correctness for each combination of ST-elevation presence, visualization type, and visualization presence conditions. Standard errors are plotted as error bars.

Table 2: og odds of success on the classification task for visualisation-present and visualisation-absent conditions (a), along with reaction time statistics (b).

(a) Log odds of success in the ST-elevation task.

Condition	Log odds of Success	Standard Error	95% CI
Visualisation-present	2.21	0.109	[1.99, 2.42]
Visualisation-absent	0.82	0.117	[0.59, 1.05]

(b) Mean reaction times, standard errors, and 95% CIs. On average, participants were 0.52 seconds quicker when a visualisation was present.

Condition	Mean RT (s)	Standard Error	95% CI
Visualisation-present	3.84	0.114	[3.61, 4.06]
Visualisation-absent	4.36	0.130	[4.1, 4.61]

Our first hypothesis therefore received support in this experiment. We used the **emmeans** package (version 1.11.2-8 [84]) to interpret the probabilities of participants choosing correctly on the ST-elevation classification task; Table 2a displays these statistics, along with standard errors and 95% confidence intervals (CI). Figure 5 shows violin plots of mean correctness separately for visualisation-present and visualisation-absent conditions.

To investigate the second hypothesis, that participants would make quicker judgements of ST-elevation when visualisations were present versus when they were absent, we built a model whereby participant judgement speed was predicted by the presence or absence of a visualisation. This model featured random intercepts for items and participants. A likelihood ratio test revealed that the model including reaction time explained significantly more variance than the null model ($\chi^2(1) = 27.56, p < .001$). Table 2b shows mean reaction times for visualisation-present and visualisation-absent conditions, along with standard errors and 95% confidence intervals. Our second hypothesis therefore also received support in this experiment; participants made ST-elevation classification decisions significantly faster when a visualisation was present.

4.4. Interaction Effects between Visualisations and Presence of ST-elevation

We built a model whereby performance on the ST-elevation classification task was predicted by whether or not a particular ECG tracing actually showed

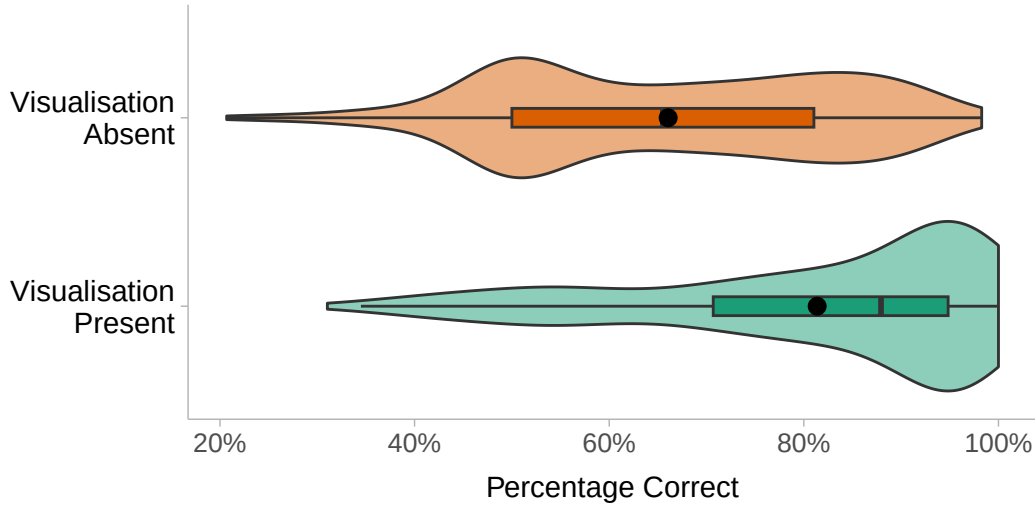


Figure 5: Percentage Correctness Distributions: mean correctness percentages for visualisation-present and visualisation-absent conditions. Boxplots show medians and interquartile ranges. Means of mean correctness percentages are shown as dots.

ST-elevation. A likelihood ratio test between this model and the null revealed no significant differences ($\chi^2(1) = 0.10$, $p = .747$), suggesting that instances of ST-elevation and no ST-elevation are processed symmetrically by participants. This model featured random intercepts for items and participants, and random slopes for participants with regard to the presence or absence of ST-elevation. To further understand how the *type* of visualisation impacts participants' decisions, we built a model whereby performance on the task was predicted by the type of visualisation employed. This model only used data from trials that featured a visualisation. A likelihood ratio test between this model and the null, again, revealed no significant differences ($\chi^2(2) = 2.64$, $p = .267$), suggesting that while participants benefitted from the presence of a visualisation, they benefitted approximately equally regardless of visualisation type.

Despite finding no evidence of fixed effects of either ST-elevation presence or visualisation type, it may be that there is an interaction present that could help explain the differences in mean correctness between visualisation types seen in Figure 4. To this end, we built a model whereby participants' performance on the ST-elevation classification task was predicted by both the type of visualisation and whether or not the ECG tracing objectively displayed ST-elevation, featuring random slopes for items with regard to the

type of visualisation used. A likelihood ratio test comparing this model to one with the fixed effects removed revealed significant differences ($\chi^2(5) = 47.71$, $p < .001$).

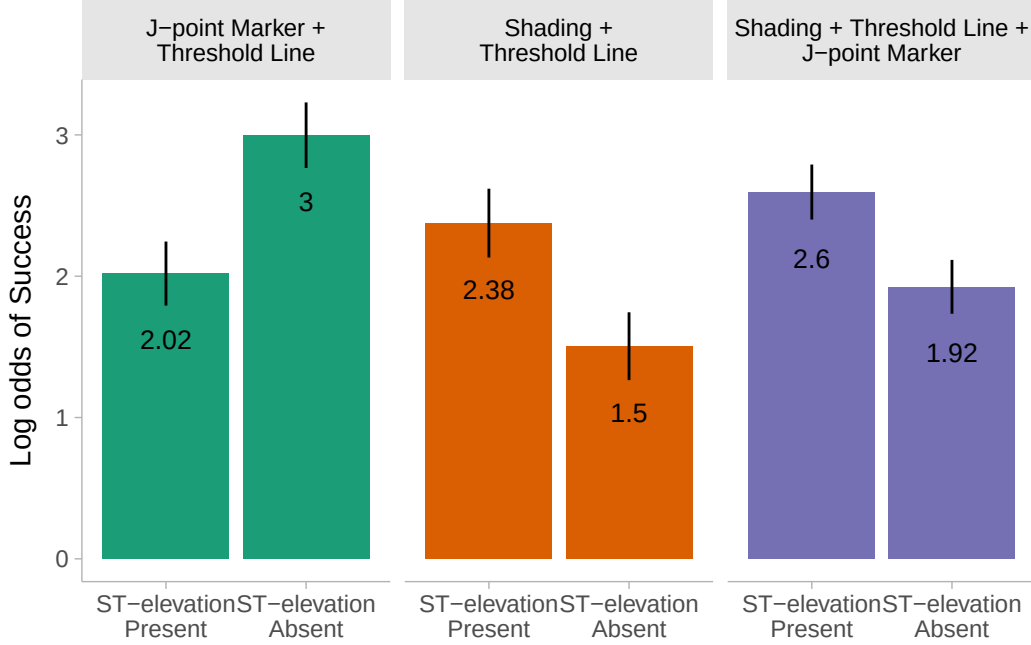


Figure 6: Comparing ST-elevation Present vs Absent: Log odds of success on the ST-elevation classification task plotted separately for each visualisation type, in both ST-elevation present and ST-elevation absent conditions. Standard errors are shown as error bars.

To explore this interaction further, we plot log odds of success for each visualisation type separately for ST-elevation present and ST-elevation absent conditions in Figure 6. Statistical testing between ST-elevation present and ST-elevation absent conditions for each visualisation was performed with the `emmeans` package and is shown in Table 3. There are significant differences between participants' performance in ST-elevation present and ST-elevation absent conditions for each visualisation type. In the *J-point Marker + Threshold Line* condition, participants performed significantly better when ST-elevation was absent, whereas in the *Shading + Threshold Line* and *Shading + Threshold Line + J-point Marker* conditions, participants performed significantly better when ST-elevation was present.

Table 3: Contrasts between ST-elevation present and ST-elevation absent conditions for each visualisation type. The p -values are adjusted using the Tukey method for multiple comparisons.

Visualisation type	Estimate	Standard Error	Z-ratio	p -value
J-point Marker + Threshold Line	-0.98	0.19	-5.03	<0.001
Shading + Threshold Line	0.87	0.29	3.00	0.033
Shading + Threshold Line + J-point Marker	0.67	0.17	4.01	<0.001

5. Discussion

Across both qualitative validation and quantitative, controlled experimental studies, we found clear evidence that visualisations based on clinical rules can meaningfully improve the identification of ST-elevation.

5.1. Trust and Transparency Through Rule-based Visualisations

Clinicians found rule-based visualisations trustworthy because they aligned with established guidelines and made their underlying logic transparent. This transparency complemented opaque automated outputs by clarifying why ST-elevation was suggested, increasing confidence and reducing cognitive load. Rather than replacing judgement, visualisations were seen as “back-up” evidence to verify or challenge interpretations – particularly important given known inconsistencies in existing CIE systems [10, 11].

5.2. J-point Localisation and the Potential for Standardisation

Clinicians identified J-point localisation as a key source of inter-rater variability and a major challenge in ECG interpretation, consistent with prior evidence [66, 65]. The J-point marker was therefore viewed as the most valuable feature of the visualisations, offering a clear, consistent anchor that reduced subjectivity. By addressing ambiguity in ST-segment assessment, it has the potential to standardise interpretation across experience levels. Clinicians emphasised its particular value in high-pressure and pre-hospital settings, where rapid, reliable identification of ST-elevation is critical, including for paramedics and first responders [2].

5.3. Educational and Workflow Benefits

Clinicians viewed the visualisations as valuable educational tools, particularly for students, junior staff, and infrequent ECG interpreters. The visualisations were perceived as aids that could help users progressively build interpretative skills, echoing the role of perceptual cues in supporting learning and reducing cognitive load [64, 85]. Participants emphasised the importance of toggling visualisations on and off, enabling learners to test independent interpretation before accessing confirmatory guidance. In practice, visual cues were seen to complement rather than replace existing CIE outputs. They clarified ambiguous automated interpretations, supporting more informed decision-making. Clinicians noted that combining textual outputs with rule-based visualisations could reduce errors, improve situational awareness, and enhance efficiency in complex cases—aligning with evidence that explainability strengthens trust and human-AI collaboration [18, 23, 24].

5.4. Visualisation Improves Classification Performance

Participants had a significantly higher log odds of success when visualisations were present compared to when they were absent, with an 81.37% correctness when a visualisation was present compared to 66.06% when the visualisation was absent. Generally, the participants in our study exhibited accuracy rates in line with those seen in clinicians [86, 87], with the caveat that the clinicians in those studies observed full 12-lead ECGs compared to the single-lead, single-heartbeat stimuli used here. For the base task – identifying ST-elevation – lay participants performed comparably to clinicians. This is in agreement with previous work [28, 29, 30], demonstrating that lay people can be trained to successfully interpret ECGs and that novel additions of visualisations help them to do so.

Participants were quicker at making classification decisions across the board when visualisations were present compared to when they were absent (see Table 2b), and for the latter, mean reaction time was still relatively small. To our knowledge this is the first experimental study in which participants (lay or otherwise) classified solely on single-lead, single-heartbeat ECG tracings. With only 4 minutes of basic training, participants were able to perform the task quickly and successfully, and provided a large amount of high-quality data. This finding is promising for future work into how lay people interpret healthcare visualisations.

5.4.1. Visualisation Type and Mechanism of Action

Each visualisation type was designed based on the clinical rules outlined in Section 2.1. Ignoring the contiguity rule, the rule for classifying ST-elevation is 0.1 mV of elevation at the J-point. The dashed threshold line at 0.1 mV was included in the visualisations to address borderline issues where low levels of ST-elevation are difficult to identify [65]. The J-point marker supported participants in locating the J-point, and the shading reduced the task load by removing the obligation to manually locate the J-point. Examples of all visualisation types, including an ECG tracing without any additional visualisation, are shown in Figure 2.

As can be seen in Figure 6, the different visualisation types interact with the presence or absence of an ST-elevation. Similar patterns of performance are observed in the *Shading + Threshold Line* and *Shading + Threshold Line + J-point Marker* conditions; these visualisations appear to have been most helpful when ST-elevation was present. The opposite pattern of results was observed in the *J-point Marker + Threshold Line* condition, where performance was best when ST-elevation was absent.

We examine the potential mechanisms behind the *Shading + Threshold Line* and *Shading + Threshold Line + J-point Marker* conditions together given the similarity in participants’ performance. In the training for both of these conditions, participants were instructed to classify the stimulus as “ST-elevation” when they could see any shading of the ST-segment. When the stimulus in question is not a boundary case, the *Shading* represents a heuristic relying on pre-attentive processing [85] to cognitively support [88] participants by removing the need to find and examine the J-point. The fact that the inclusion of the *J-point Marker* in addition to *Shading* benefits participants’ performance, may be explained by the visual anchor provided by the *J-point Marker* both for (1) boundary cases where ST-elevation is present but for which almost no shading is visible; and (2) for cases where ST-elevation is absent and thus, only the threshold line would be visible, with no shading.

While *Shading* is helpful when ST-elevation is present, the results suggest that the *J-point Marker + Threshold Line* alone is more helpful when ST-elevation is absent. As can be seen in Figure 6, participants’ log odds of success on the task was highest in the ST-elevation absent, *J-point Marker + Threshold Line* condition. In this condition, participants were told that

the green circle marked the J-point. While not providing as much cognitive support as the *Shading*, the J-point marker effectively removes the need for participants to manually locate the J-point. Removing this requirement may afford participants more cognitive resources for determining whether or not the J-point is elevated above the thresholdline. The exact mechanism behind why the *J-point marker + Threshold Line* works so well for ST-elevation absent conditions is unclear, considering the visualisations presented are in fact the same as in the *Shading + Threshold Line + J-point Marker* conditions (since there is no shading). Proximity of the J-point marker to the threshold line may play a role in classification accuracy. This distance may be larger when ST-elevation is absent, making the visual cue provided by the J-point marker more pronounced, and thus easier to identify. However, participants from the *Shading + Threshold Line + J-point Marker* may rely more heavily on visual cues provided by the shading than on the concurrently available J-point marker. Prior exposure to shading may prime participants to perceive (i.e. hallucinate) shaded regions in borderline cases.

Despite showing identical stimuli for the visualisation absent conditions across the three participation groups, there is a variation in correctness (see Figure 4), suggesting potential cognitive artefacts due to exposures to the different visualisation types [89] whose explanation will require future work.

6. Conclusion

Our study demonstrates that rule-based, explainable ECG visualisations can significantly improve the identification of ST-segment elevation across lay users and provide useful information for clinicians to support their decision-making. Practising clinicians confirmed the clinical relevance of our visualisations. They consistently valued the transparency of rule-based cues, the reliability of explicit J-point marking, and the potential for these visualisations to complement, rather than replace, existing CIE systems. Lay participants – after only brief training – achieved higher accuracy and faster response times when visualisations were present, showing that intuitive, guideline-aligned cues can effectively support ECG interpretation even among non-experts.

Looking ahead, future work should evaluate these visualisations within real clinical workflows and full 12-lead contexts, assessing their impact in emergency departments, pre-hospital care, and EMS decision-making. Integrating these visual cues into automated ECG interpretation systems may

also enhance explainability and clinicians’ trust, particularly as ECG-enabled wearables and AI-driven monitoring become more widespread. Further exploration using eye-tracking, psychophysics, and studies with patient-generated ECG data will help clarify the mechanisms behind visual cue effectiveness and guide refinement of interface designs.

Our findings indicate that simple, transparent, rule-based visualisations can reduce interpretive variability, improve diagnostic confidence, and support both learning and real-world decision-making. By directly addressing known challenges such as J-point localisation and the explainability gap in automated ECG interpretation, these visual aids offer a promising avenue towards safer, more interpretable, and more accessible ECG diagnostics.

Funding Sources

This work was funded by the Engineering and Physical Sciences Research Council (EPSRC) under grant number EP/X02945X/1.

CRedit authorship contribution statement

Lukas Hughes-Noehrer: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Software, Supervision, Visualization, Writing – original draft, Writing – review & editing. **Gabriel Strain:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Visualization, Writing – original draft, Writing – review & editing. **Adina Rahim:** Data curation, Software, Validation. **James Sinnott:** Data curation, Software, Supervision, Validation. **Jonathan Carlton:** Data curation, Methodology, Software. **Crystal Wu:** Data curation, Software, Validation, Visualization, Writing – review & editing. **Rick Body:** Conceptualization, Funding acquisition, Supervision, Validation, Writing – review & editing. **Caroline Jay:** Conceptualization, Funding acquisition, Methodology, Project administration, Resources, Supervision, Validation, Writing – review & editing.

Declaration of competing interest

The authors have no competing interests to declare.

Acknowledgements

We want to thank all clinicians and critical friends of the project for their continued input as part of our Responsible Research and Innovation stakeholder engagement.

Data availability

The ECG-X project website is available at www.ecgx.co.uk. This study followed open and reproducible research practices, with hypotheses and analysis plans pre-registered on the Open Science Framework (<https://osf.io/v6tjp>). The algorithm to generate ECG images with visualisations is publicly accessible on GitHub (<https://doi.org/10.5281/zenodo.20214308>), alongside a repository containing all data and analysis code (https://github.com/ECG-X/increasing_the_accuracy_of_ST-segment_elevation), including instructions to build a Docker environment enabling full replication of analyses, figures, and the manuscript. Supplementary materials for the clinical validation study including Ethics Outcome Letters, participant information sheet and consent form, and interview transcripts, are hosted on Figshare (<https://doi.org/10.48420/31042243>). Experimental code, materials, and procedures are available on GitLab as three separate experiments (https://gitlab.pavlovia.org/Strain/ecg_x_jt, https://gitlab.pavlovia.org/Strain/ecg_x_st, and https://gitlab.pavlovia.org/Strain/ecg_x_stj).

Appendix A. Algorithm Development

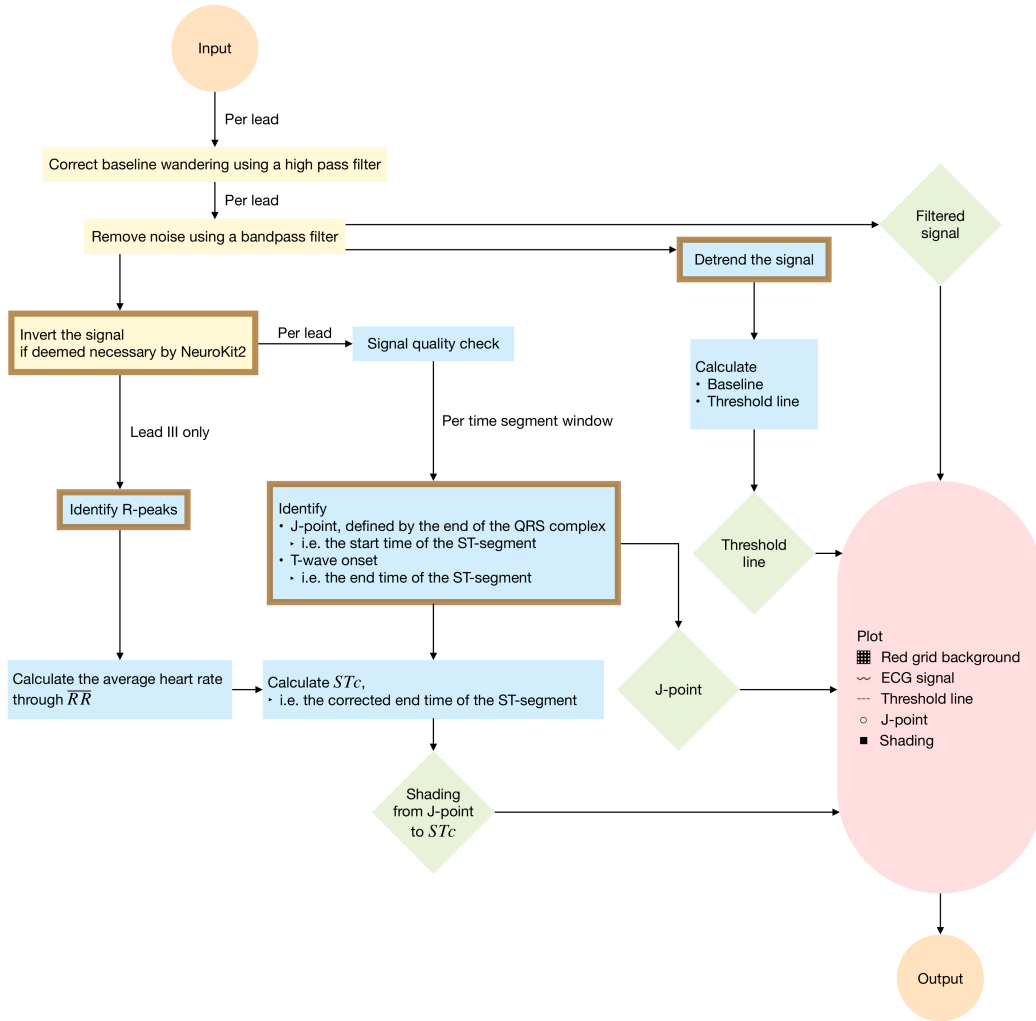


Figure A.1: Computational pipeline for ST-segment features detection and visualisation: Data preprocessing (in yellow) includes applying a high pass and a bandpass filter. Some processing is carried out in segments, each with a duration of the segment time window defined in configuration. Steps completed utilising NeuroKit2 functions (circled in brown), identify features relevant to ST-elevations (in blue). The four distinctive visualisations (in green) are plotted on a red grid background with the ECG signal (in pink).

A flowchart depicting our full pipeline is shown in Figure A.1. The codebase

of our full pipeline¹ is openly available on Zenodo under a dual-licensing model: released for non-commercial use under CC BY-NC-SA 4.0, with alternative commercial licensing available upon request.

Appendix A.1. Technical Assumptions

Our algorithm was intentionally designed to produce novel, foundational, human-understandable visualisations to facilitate the interpretation of ECGs. The following assumptions were made:

- Given the complexity and variety of raw ECG signals, we acknowledge that our pipeline is tuned to suit the PTB Diagnostics database [90] specifically, containing clinically labelled, high-quality signals.
- We focus on the human-algorithm interaction aspect of algorithm development, using well-known, standard open-source processing packages and algorithms for ECG signal processing.
- The proposed visualisation pipeline prioritises perceptual clarity and interpretability over optimal representation of physiological states. Our contribution lies in novel *clinical rule-based visualisations* intended to standardise visual representation rather than to model underlying electrophysiology with maximal accuracy. This design choice reflects the development focus on human interpretation and decision support rather than physiological measurement accuracy (a common trade-off in CIE design rather than a limitation specific to our method).

Appendix A.2. Data Preparation

Raw ECG data were acquired from the clinically-validated PTB Diagnostic ECG Database [90], which is openly available and can be downloaded from PhysioNet. Prior to any signal processing, the raw data were converted from WaveForm DataBase (WFDB) format to CSV for ingestion into our *ST-segment feature detection and visualisation* pipeline. From the 549 available datasets downloaded and converted to CSV, we grouped them into three categories based on the diagnosis (i.e. reason for admission) provided by the

¹<https://doi.org/10.5281/zenodo.20214308>

database. There were 80 ECG recordings classified as *healthy control*, 368 *myocardial infarction*, and the *other* remaining 101 includes other heart conditions (e.g., cardiomyopathy, bundle branch block, etc.) and miscellaneous. We selected data from the *healthy control* and the *myocardial infarction* group as input, as they most likely contain signals without and with ST-elevation, respectively. Each recording is represented by a CSV file containing data, including all of the leads provided by the database, from the same recording session. An ST-segment lasts approximately 80 ms to 120 ms [91], and since the ECG datasets have varying durations, we configured our pipeline to sample the first 30 s from each recorded signal. Each signal therefore has approximately 39 potential ST-segments (based on the mean real-world heart rate of 79.1 beats per minute (bpm) [92]) available for analysis and can be visually annotated with our visualisations.

Appendix A.3. Data Preprocessing

We utilised SciPy [93] and NeuroKit2 [94], two openly available Python libraries, for signal preprocessing tasks. The first two steps of preprocessing aimed to reduce noise while preserving the signal morphology.

Artefacts caused by respiration, movement, electrode impedance, and electrical interference [95] distort ECG signal quality [96] and hinder feature detection relevant to ST-segment interpretation and analysis [97]. To correct baseline wander, we applied a Butterworth [98] high pass filter, chosen for its accuracy and computational efficiency in medical applications [97]. A cut-off frequency at 0.5 Hertz (Hz) [96, 99, 100] removed low frequency noise and shifted the signal vertically to centre around zero. Then, a second Butterworth bandpass filter (0.5 Hz to 50 Hz) was used to reduce high frequency noise due to electrical (e.g., powerline) and myoelectric interferences [96, 99, 101], improving signal clarity. Deflections from the raw signals are preserved in the filtered signals, which form the basis of the ECG tracings (with and without visualisations) used in the online experiment with lay participants.

Our third step utilised NeuroKit2 to identify and invert any *inverted signals* (an artefact due to electrode placement during the recording session or pathological causes [102, 103, 104]). Signals identified to have predominantly upright features with positive deflection were kept unchanged. This is to support detecting ECG waveform features, specifically those relevant to an ST-segment.

Appendix A.4. Pipeline Configuration and Data Quality Check

The pipeline is built to support the 12 conventional leads routinely used in clinical practice [105], where the leads are selectable through configuration for processing. The algorithm is also capable of applying the threshold line based on the ST-elevation criteria depending on the age and sex of the patient. As our online experiment with lay participants excluded evaluating leads V2 and V3 (see Section 3.4), the pipeline was configured accordingly.

ECG signals from the PTB Diagnostic ECG database were measured within the range of ± 16.384 mV [90], yet ECG amplitudes are generally below 5 mV [106]. For quality control and to ensure the output ECG images have a consistent height regardless of signal amplitude, we adopted a conservative approach by configuring the pipeline to discard datasets with maximum amplitudes outside ± 3 mV, as well as signals shorter than 30 s. Excluding leads V2 and V3, we determined 72.5% (58 out of 80 recordings) and 64.1% (236 out of 368 recordings) of the datasets from the *healthy control* and the *myocardial* group, respectively, have maximum amplitudes within ± 3 mV. In addition, to streamline the remaining steps, we partitioned the 30 s signal by a configurable time segment window (e.g., 5 s) to process. For the final output, the number of ECG images produced per signal for a lead is dependent on the specified time segment window and the total duration of a signal sampled for input.

Appendix A.5. Feature Detection

Recognising ST-elevation requires analysing the signal against the clinically defined threshold level, an electrical potential displacement from the signal baseline, and can be computed once the baseline is established. The baseline, also known as the isoelectric line, represents net zero electrical potential [107]. As our filtered signal already centre vertically around zero, we utilised NeuroKit2 to remove the linear trend and define the resulting signal’s mean as our baseline. Although it approximates the isoelectric line, we acknowledge that our custom baseline may be skewed towards the side with greater cumulative deflection in the waveform.

An ST-segment, shown in Figure 1, is defined by the interval from the end of a QRS complex to the start of the following T-wave. It is typically an isoelectric period near the baseline or may slope slightly upward in a normal state [105, 107]. We utilised NeuroKit2 to identify prominent features of each

signal, prepared by the steps discussed in Appendix A.3. Implementation via NeuroKit2 includes the Martinez et al. method [108] to identify R-peaks and the discrete wavelet transform (DWT) method to identify onsets and offsets of relevant waves, including the QRS complex and the T-wave, to determine an ST-segment and its J-point. R-peaks are typically the most prominent feature in an ECG tracing and can be used to compute the average heart rate. We achieved this by taking the mean of the interval between two consecutive R-peaks ($\overline{RR}$) using data from lead III.

Appendix A.6. Algorithm Output

Based on the ECG features detected by our algorithm pipeline, visualisations were applied (i.e. visually annotated) on top of the filtered ECG signal, where noises have been removed. The final output consists of ECG images in a style familiar to clinicians. Each small grid of the background is 1 mm. The time axis was configured to 25 mm/s and the electrical potential axis was set to 10 mm/mV.

As our custom baselines approximate 0 mV, the ECG signals essentially align vertically to the centre of the red grid background. Both the *Threshold Line* and the *J-point Marker* for assessing ST-elevation are applied based on the computed threshold value and the determined J-point location. *Shading* of an ST-segment is achieved by applying a polygon patch from the time of the J-point to the cut-off time indicated by *STc*, *only if* the value of the signal within this time frame is larger than the computed threshold, akin to area under the curve visually.

Appendix B. Clinician Demographics

Table B.4: Participant demographics split by specialty and training stage

Participant Number (P)	Specialty	Training Stage
1	EM	Consultant
2	Cardio	ST4+
3	An&CC	Consultant
4	EM	ST4+
5	EM	SAS
6	Cardio	ST4+

Notes: Anaesthesia (An); Critical Care (CC); Emergency Medicine (EM); Cardio (Cardiology). A comprehensive list outlining UK medical doctor titles and training stages can be found in [109].

Appendix C. Experimental trial example

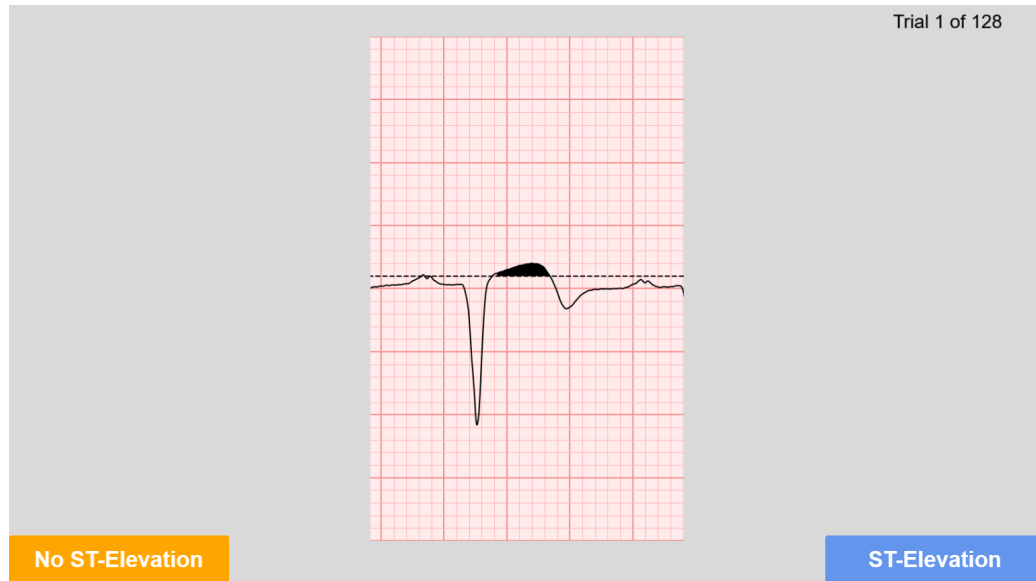


Figure A.2: Example Experimental Trial: example trial in the *Shading + Threshold Line* condition. This ECG visualisation shows ST-elevation.

References

- [1] I. C. Moraes-Silva, B. Rodrigues, H. J. Coelho-Junior, D. J. Feriani, M.-C. Irigoyen, Myocardial Infarction and Exercise Training: Evidence from Basic Science, *Advances in Experimental Medicine and Biology* 999 (2017) 139–153. doi:10.1007/978-981-10-4307-9_9.
- [2] B. Ibanez, S. James, S. Agewall, M. J. Antunes, C. Bucciarelli-Ducci, H. Bueno, A. L. P. Caforio, F. Crea, J. A. Goudevenos, S. Halvorsen, G. Hindricks, A. Kastrati, M. J. Lenzen, E. Prescott, M. Roffi, M. Valgimigli, C. Varenhorst, P. Vranckx, P. Widimský, ESC Scientific Document Group, 2017 ESC Guidelines for the management of acute myocardial infarction in patients presenting with ST-segment elevation: The Task Force for the management of acute myocardial infarction in patients presenting with ST-segment elevation of the European Society of Cardiology (ESC), *European Heart Journal* 39 (2) (2018) 119–177. doi:10.1093/eurheartj/ehx393.

- [3] J. Hampton, J. Hampton, D. Adlam, The ECG Made Easy E-Book: The ECG Made Easy E-Book, Elsevier Health Sciences, 2024.
- [4] S. M. Salerno, P. C. Alguire, H. S. Waxman, Competency in interpretation of 12-lead electrocardiograms: A summary and appraisal of published evidence, *Annals of Internal Medicine* 138 (9) (2003) 751–760.
- [5] D. A. Cook, S.-Y. Oh, M. V. Pusic, Accuracy of Physicians’ Electrocardiogram Interpretations: A Systematic Review and Meta-analysis, *JAMA Internal Medicine* 180 (11) (2020) 1461. doi:10.1001/jamainternmed.2020.3989.
- [6] N. Rafie, A. H. Kashou, P. A. Noseworthy, ECG Interpretation: Clinical Relevance, Challenges, and Advances, *Hearts* 2 (4) (2021) 505–513. doi:10.3390/hearts2040039.
- [7] NHS Digital, NHS Workforce Statistics - December 2020 (Including selected provisional statistics for January 2021), <https://digital.nhs.uk/data-and-information/publications/statistical/nhs-workforce-statistics/december-2020>, accessed 14 August 2025 (2020 and 2021).
- [8] P. Mele, Improving electrocardiogram interpretation in the clinical setting, *Journal of Electrocardiology* 41 (5) (2008) 438–439. doi:10.1016/j.jelectrocard.2008.04.003.
- [9] P. W. Macfarlane, J. Kennedy, Automated ECG Interpretation—A Brief History from High Expectations to Deepest Networks, *Hearts* 2 (4) (2021) 433–448, number: 4 Publisher: Multidisciplinary Digital Publishing Institute. doi:10.3390/hearts2040034.
- [10] H. Smulyan, The Computerized ECG: Friend and Foe, *The American Journal of Medicine* 132 (2) (2019) 153–160. doi:10.1016/j.amjmed.2018.08.025.
- [11] J. Schläpfer, H. J. Wellens, Computer-Interpreted Electrocardiograms, *Journal of the American College of Cardiology* 70 (9) (2017) 1183–1192, publisher: American College of Cardiology Foundation. doi:10.1016/j.jacc.2017.07.723.

- [12] S. Gupta, A. H. Kashou, R. Herman, S. Smith, A. May, A. G. M. Echeverri, M. Del Sueldo, A. C. Berni, J. Farina, S. Garcia-Zamora, A. Baranchuk, Computer-Interpreted Electrocardiograms: Impact on Cardiology Practice, *International Journal of Cardiovascular Sciences* 37 (2024) e20240079, publisher: Sociedade Brasileira de Cardiologia. doi:10.36660/ijcs.20240079.
- [13] S. K. Gill, A. Karwath, H.-W. Uh, V. R. Cardoso, Z. Gu, A. Barsky, L. Slater, A. Acharjee, J. Duan, L. Dall’Olio, S. el Bouhaddani, S. Chernbumroong, M. Stanbury, S. Haynes, F. W. Asselbergs, D. E. Grobbee, M. J. C. Eijkemans, G. V. Gkoutos, D. Kotecha, Artificial intelligence to enhance clinical value across the spectrum of cardiovascular healthcare, *European Heart Journal* 44 (9) (2023) 713–725. doi:10.1093/eurheartj/ehac758.
- [14] A. Lyon, A. Mincholé, J. P. Martínez, P. Laguna, B. Rodriguez, Computational techniques for ECG analysis and interpretation in light of their contribution to medical advances, *Journal of The Royal Society Interface* 15 (138) (2018) 20170821, publisher: Royal Society. doi:10.1098/rsif.2017.0821.
- [15] J. C. L. Himmelreich, R. E. Harskamp, Diagnostic accuracy of the PMcardio smartphone application for artificial intelligence-based interpretation of electrocardiograms in primary care (AMSTELHEART-1), *Cardiovascular Digital Health Journal* 4 (3) (2023) 80–90. doi:10.1016/j.cvdhj.2023.03.002.
- [16] S. Khurshid, Clinical perspectives on the adoption of the artificial intelligence-enabled electrocardiogram, *Journal of Electrocardiology* 81 (2023) 142–145. doi:10.1016/j.jelectrocard.2023.08.014.
- [17] K. C. Siontis, P. A. Noseworthy, Z. I. Attia, P. A. Friedman, Artificial intelligence-enhanced electrocardiography in cardiovascular disease management, *Nature Reviews Cardiology* 18 (7) (2021) 465–478, publisher: Nature Publishing Group. doi:10.1038/s41569-020-00503-2.
- [18] Y. Elul, A. A. Rosenberg, A. Schuster, A. M. Bronstein, Y. Yaniv, Meeting the unmet needs of clinicians from AI systems showcased for cardiology with deep-learning-based ECG analysis, *Proceedings of the National Academy of Sciences* 118 (24) (2021) e2020620118,

publisher: Proceedings of the National Academy of Sciences. doi: 10.1073/pnas.2020620118.

- [19] C. T. Chung, S. Lee, E. King, T. Liu, A. A. Armoundas, G. Bazoukis, G. Tse, Clinical significance, challenges and limitations in using artificial intelligence for electrocardiography-based diagnosis, *International Journal of Arrhythmia* 23 (1) (2022) 24. doi:10.1186/s42444-022-00075-x.
- [20] P. Rajpurkar, E. Chen, O. Banerjee, E. J. Topol, AI in health and medicine, *Nature Medicine* 28 (1) (2022) 31–38, publisher: Nature Publishing Group. doi:10.1038/s41591-021-01614-0.
- [21] A. Schepart, A. Burton, L. Durkin, A. Fuller, E. Charap, R. Bhambri, F. S. Ahmad, Artificial intelligence-enabled tools in cardiovascular medicine: A survey of current use, perceptions, and challenges, *Cardiovascular Digital Health Journal* 4 (3) (2023) 101–110, publisher: Elsevier. doi:10.1016/j.cvdhj.2023.04.003.
- [22] J. Stewart, S. Freeman, E. Eroglu, N. Dumitrascu, J. Lu, A. Goudie, P. Sprivulis, H. Akhlaghi, V. Tran, F. Sanfilippo, A. Celenza, M. Than, D. Fatovich, K. Walker, G. Dwivedi, Attitudes towards artificial intelligence in emergency medicine, *Emergency Medicine Australasia* 36 (2) (2024) 252–265, _eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1111/1742-6723.14345>. doi:10.1111/1742-6723.14345.
- [23] E. R. Burgess, I. Jankovic, M. Austin, N. Cai, A. Kapuścińska, S. Currie, J. M. Overhage, E. S. Poole, J. Kaye, Healthcare AI Treatment Decision Support: Design Principles to Enhance Clinician Adoption and Trust, in: *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*, ACM, Hamburg Germany, 2023, pp. 1–19. doi: 10.1145/3544548.3581251.
- [24] O. Asan, A. E. Bayrak, A. Choudhury, Artificial Intelligence and Human Trust in Healthcare: Focus on Clinicians, *Journal of Medical Internet Research* 22 (6) (2020) e15154, company: Journal of Medical Internet Research Distributor: Journal of Medical Internet Research Institution: Journal of Medical Internet Research Label: Journal of Medical Internet

Research Publisher: JMIR Publications Inc., Toronto, Canada. doi: 10.2196/15154.

- [25] L. Hughes-Noehrer, L. Channer, G. Strain, G. Yates, R. Body, C. Jay, Contrasting attitudes towards current and future artificial intelligence applications for computerised interpretation of electrocardiograms: A clinical stakeholder interview study, *JAMIA Open* 8 (4) (2025) ooaf071. doi:10.1093/jamiaopen/ooaf071.
- [26] L. Hughes-Noehrer, A. Alahmadi, L. Channer, A. Rahim, R. Body, C. Jay, Attitudes of clinicians to a ‘human-like’ explainable ai based on pseudo-colouring of ecgs that exposes life-threatening anomalies, *Journal of Electrocardiology* 85 (2024) 4. doi:<https://doi.org/10.1016/j.jelectrocard.2024.06.009>.
URL <https://www.sciencedirect.com/science/article/pii/S0022073624001833>
- [27] S. V. Rao, M. L. O’Donoghue, M. Ruel, T. Rab, J. E. Tamis-Holland, J. H. Alexander, U. Baber, H. Baker, M. G. Cohen, M. Cruz-Ruiz, L. L. Davis, J. A. de Lemos, T. A. DeWald, I. Y. Elgendy, D. N. Feldman, A. Goyal, I. Isiadinso, V. Menon, D. A. Morrow, D. Mukherjee, E. Platz, S. B. Promes, S. Sandner, Y. Sandoval, R. Schunder, B. Shah, J. P. Stopyra, A. W. Talbot, P. R. Taub, M. S. Williams, 2025 ACC/AHA/ACEP/NAEMSP/SCAI Guideline for the Management of Patients With Acute Coronary Syndromes: A Report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines, *Circulation* 151 (13) (2025) e771–e862. doi:10.1161/CIR.0000000000001309.
- [28] A. Alahmadi, A. Davies, M. Vigo, C. Jay, Can laypeople identify a drug-induced QT interval prolongation? A psychophysical and eye-tracking experiment examining the ability of nonexperts to interpret an ECG, *Journal of the American Medical Informatics Association* 26 (5) (2019) 404–411. doi:10.1093/jamia/ocy183.
- [29] A. Alahmadi, A. Davies, J. Royle, M. Vigo, C. Jay, Evaluating the Impact of Pseudo-Colour and Coordinate System on the Detection of Medication-induced ECG Changes, in: *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, ACM, Glasgow Scotland Uk, 2019, pp. 1–13. doi:10.1145/3290605.3300353.

- [30] A. Alahmadi, A. Davies, M. Vigo, C. Jay, Pseudo-colouring an ECG enables lay people to detect QT-interval prolongation regardless of heart rate, *PLOS ONE* 15 (8) (2020) e0237854. doi:10.1371/journal.pone.0237854.
- [31] A. Alahmadi, A. Davies, J. Royle, L. Goodwin, K. Cresswell, Z. Arain, M. Vigo, C. Jay, An explainable algorithm for detecting drug-induced QT-prolongation at risk of torsades de pointes (TdP) regardless of heart rate and T-wave morphology, *Computers in Biology and Medicine* 131 (2021) 104281. doi:10.1016/j.combiomed.2021.104281.
- [32] A. Alahmadi, A. Davies, M. Vigo, C. Jay, Personalized, intuitive & visual QT-prolongation monitoring using patient-specific QTc threshold with pseudo-coloring and explainable AI, *Journal of Electrocardiology* 81 (2023) 218–223. doi:10.1016/j.jelectrocard.2023.09.012.
- [33] N. Ojha, A. S. Dhamoon, Myocardial Infarction, in: *StatPearls*, StatPearls Publishing, Treasure Island (FL), 2023.
- [34] K. Thygesen, J. S. Alpert, A. S. Jaffe, B. R. Chaitman, J. J. Bax, D. A. Morrow, H. D. White, ESC Scientific Document Group, Fourth universal definition of myocardial infarction (2018), *European Heart Journal* 40 (3) (2019) 237–269. doi:10.1093/eurheartj/ehy462.
- [35] H. Akbar, S. Mountfort, Acute ST-Segment Elevation Myocardial Infarction (STEMI), in: *StatPearls*, StatPearls Publishing, Treasure Island (FL), 2024.
- [36] I. Pascual, D. Hernandez-Vaquero, M. Almendarez, R. Lorca, A. Escalera, R. Díaz, A. Alperi, M. Carnero, J. Silva, C. Morís, P. Avanzas, Observed and Expected Survival in Men and Women after Suffering a STEMI, *Journal of Clinical Medicine* 9 (4) (2020) 1174. doi:10.3390/jcm9041174.
- [37] F. Pedersen, V. Butrymovich, H. Kelbæk, K. Wachtell, S. Helqvist, J. Kastrup, L. Holmvang, P. Clemmensen, T. Engstrøm, P. Grande, K. Saunamäki, E. Jørgensen, Short- and Long-Term Cause of Death in Patients Treated With Primary PCI for STEMI, *JACC* 64 (20) (2014) 2101–2108. doi:10.1016/j.jacc.2014.08.037.

- [38] C. P. Gale, V. Allan, B. A. Cattle, A. S. Hall, R. M. West, A. Timmis, H. H. Gray, J. Deanfield, K. a. A. Fox, R. Feltbower, Trends in hospital treatments, including revascularisation, following acute myocardial infarction, 2003–2010: A multilevel and relative survival analysis for the National Institute for Cardiovascular Outcomes Research (NICOR), *Heart* 100 (7) (2014) 582–589. doi:10.1136/heartjnl-2013-304517.
- [39] W. Wang, D. R. Thompson, C. F. Ski, M. Liu, Health-related quality of life and its associated factors in Chinese myocardial infarction patients, *European Journal of Preventive Cardiology* 21 (3) (2014) 321–329. doi:10.1177/2047487312454757.
- [40] K. Kang, L. Gholizadeh, S. C. Inglis, H.-R. Han, Interventions that improve health-related quality of life in patients with myocardial infarction, *Quality of Life Research* 25 (11) (2016) 2725–2737. doi:10.1007/s11136-016-1401-8.
- [41] M. Bahall, K. Khan, Quality of life of patients with first-time AMI: A descriptive study, *Health and Quality of Life Outcomes* 16 (1) (2018) 32. doi:10.1186/s12955-018-0860-8.
- [42] P. A. Friedman, The electrocardiogram at 100 years: History and future, *Circulation* 149 (6) (2024) 411–413.
- [43] H. K. Chiang, C. W. Chu, G. Y. Chen, C. D. Kuo, A new 3-D display method for 12-lead ECG, *IEEE transactions on bio-medical engineering* 48 (10) (2001) 1195–1202. doi:10.1109/10.951523.
- [44] J. Heo, J. J. Lee, S. Kwon, B. Kim, S. O. Hwang, Y. R. Yoon, A novel method for detecting ST segment elevation myocardial infarction on a 12-lead electrocardiogram with a three-dimensional display, *Biomedical Signal Processing and Control* 56 (2020) 101700. doi:10.1016/j.bspc.2019.101700.
- [45] R. Fang, C.-C. Lu, C.-T. Chuang, W.-H. Chang, A visually interpretable detection method combines 3-D ECG with a multi-VGG neural network for myocardial infarction identification, *Computer Methods and Programs in Biomedicine* 219 (2022) 106762. doi:10.1016/j.cmpb.2022.106762.

- [46] G. Bortolan, 3D ECG display with deep learning approach for identification of cardiac abnormalities from a variable number of leads, *Physiological Measurement* 44 (2) (2023) 024002. doi:10.1088/1361-6579/acb4dc.
- [47] A. Perron, T. Lim, U. Pahlm-Webb, G. S. Wagner, O. Pahlm, Maximal increase in sensitivity with minimal loss of specificity for diagnosis of acute coronary occlusion achieved by sequentially adding leads from the 24-lead electrocardiogram to the orderly sequenced 12-lead electrocardiogram, *Journal of Electrocardiology* 40 (6) (2007) 463–469. doi:10.1016/j.jelectrocard.2007.07.002.
- [48] K. Xu, S. Guo, N. Cao, D. Gotz, A. Xu, H. Qu, Z. Yao, Y. Chen, ECGLens: Interactive Visual Exploration of Large Scale ECG Data for Arrhythmia Detection, in: *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems, CHI '18*, Association for Computing Machinery, New York, NY, USA, 2018, pp. 1–12. doi:10.1145/3173574.3174237.
- [49] GE. Burch, JA. Abildskov, JA. Cronvich, Vectorcardiography, *Circulation* 8 (4) (1953) 605–613.
- [50] G. E. Burch, The history of vectorcardiography, *Medical History* 29 (S5) (1985) 103–131.
- [51] A. R. P. Riera, A. H. Uchida, C. F. Filho, A. Meneghini, C. Ferreira, E. Schapacknik, S. Dubner, P. Moffa, Significance of Vectorcardiogram in the Cardiological Diagnosis of the 21st Century, *Clinical Cardiology* 30 (7) (2007) 319–323. doi:10.1002/c1c.14.
- [52] R. Jaros, R. Martinek, L. Danys, Comparison of Different Electrocardiography with Vectorcardiography Transformations, *Sensors* 19 (14) (2019) 3072. doi:10.3390/s19143072.
- [53] R. L. Lux, C. R. Smith, R. F. Wyatt, J. A. Abildskov, Limited lead selection for estimation of body surface potential maps in electrocardiography, *IEEE transactions on bio-medical engineering* 25 (3) (1978) 270–276. doi:10.1109/TBME.1978.326332.
- [54] B. Taccardi, B. B. Punske, R. L. Lux, R. S. MacLEOD, P. R. Ershler, T. J. Dustman, Y. Vyhmeister, Useful Lessons from Body Surface

- Mapping, *Journal of Cardiovascular Electrophysiology* 9 (7) (1998) 773–786. doi:10.1111/j.1540-8167.1998.tb00965.x.
- [55] R. R. Bond, D. D. Finlay, C. D. Nugent, G. Moore, D. Guldenring, Methods for presenting and visualising electrocardiographic data: From temporal signals to spatial imaging, *Journal of Electrocardiology* 46 (3) (2013) 182–196. doi:10.1016/j.jelectrocard.2013.01.008.
 - [56] R. T. Sutton, D. Pincock, D. C. Baumgart, D. C. Sadowski, R. N. Fedorak, K. I. Kroeker, An overview of clinical decision support systems: benefits, risks, and strategies for success, *npj Digital Medicine* 3 (1) (2020) 17. doi:10.1038/s41746-020-0221-y.
URL <https://www.nature.com/articles/s41746-020-0221-y>
 - [57] W. Aigner, K. Kaiser, S. Miksch, Chapter 8: Visualization methods to support guideline-based care management, *Computer-Based Medical Guidelines and Protocols: A Primer and Current Trends* Volume 139 (2008).
 - [58] A. Ledesma, N. Bidargaddi, J. Strobel, G. Schrader, H. Nieminen, I. Korhonen, M. Ermes, Health timeline: an insight-based study of a timeline visualization of clinical data, *BMC Medical Informatics and Decision Making* 19 (1) (2019) 170. doi:10.1186/s12911-019-0885-x.
URL <https://bmcmmedinformdecismak.biomedcentral.com/articles/10.1186/s12911-019-0885-x>
 - [59] R. A. Dixit, S. Hurst, K. T. Adams, C. Boxley, K. Lysen-Hendershot, S. S. Bennett, E. Booker, R. M. Ratwani, Rapid development of visualization dashboards to enhance situation awareness of COVID-19 telehealth initiatives at a multihospital healthcare system, *Journal of the American Medical Informatics Association* 27 (9) (2020) 1456–1461. doi:10.1093/jamia/ocaa161.
 - [60] C. Ware, *Information Visualization: Perception for Design*, 3rd Edition, Morgan Kaufmann, Boston, MA, 2012.
 - [61] M. Sedlmair, M. Meyer, T. Munzner, Design Study Methodology: Reflections from the Trenches and the Stacks, *IEEE Transactions on Visualization and Computer Graphics* 18 (12) (2012) 2431–2440.

doi:10.1109/TVCG.2012.213.

URL <http://ieeexplore.ieee.org/document/6327248/>

- [62] A. Arcia, N. C. Benda, D. T. Y. Wu, Advancing the science of visualization of health data for lay audiences, *Journal of the American Medical Informatics Association* 31 (2) (2024) 283–288. doi:10.1093/jamia/ocad255.
- [63] J. S. Ancker, N. C. Benda, B. J. Zikmund-Fisher, Do you want to promote recall, perceptions, or behavior? The best data visualization depends on the communication goal, *Journal of the American Medical Informatics Association* 31 (2) (2024) 525–530. doi:10.1093/jamia/ocad137.
- [64] F. Paas, J. J. G. van Merriënboer, Cognitive-Load Theory: Methods to Manage Working Memory Load in the Learning of Complex Tasks, *Current Directions in Psychological Science* 29 (4) (2020) 394–398. doi:10.1177/0963721420922183.
- [65] B. F. Erling, A. D. Perron, W. J. Brady, Disagreement in the interpretation of electrocardiographic ST segment elevation: A source of error for emergency physicians?, *The American Journal of Emergency Medicine* 22 (2) (2004) 65–70. doi:10.1016/j.ajem.2003.12.004.
- [66] S. D. Carley, What’s the point of ST elevation?, *Emergency Medicine Journal* 19 (2) (2002) 126–128. doi:10.1136/emj.19.2.126.
- [67] S. Man, C. C. Ter Haar, M. C. De Jongh, A. C. Maan, M. J. Schalijs, C. A. Swenne, Position of ST-deviation measurements relative to the J-point: Impact for ischemia detection, *Journal of Electrocardiology* 50 (1) (2017) 82–89. doi:10.1016/j.jelectrocard.2016.10.012.
URL <https://linkinghub.elsevier.com/retrieve/pii/S0022073616302825>
- [68] T. Lindow, J. Wiiala, J. Lundager Forberg, A. Lassen, M. Brabrand, P. Platonov, U. Ekelund, Optimal measuring point for ST deviation in chest pain patients with possible acute coronary syndrome, *Journal of Electrocardiology* 58 (2020) 165–170. doi:10.1016/j.jelectrocard.2019.12.012.
URL <https://linkinghub.elsevier.com/retrieve/pii/S0022073619308325>

- [69] H. Bazett, An analysis of the time-relations of electrocardiograms, *Heart* 7 (1920) 353–370.
- [70] J. Mason, *Qualitative Researching*, 3rd Edition, SAGE, London, 2018.
- [71] U.S. Department of Health and Human Services, Food and Drug Administration, Center for Devices and Radiological Health, Office of Device Evaluation, Applying human factors and usability engineering to medical devices: Guidance for industry and food and drug administration staff, <https://www.fda.gov/regulatory-information/search-fda-guidance-documents/applying-human-factors-and-usability-engineering-medical-devices>, last accessed 14 January 2026 (February 2016).
- [72] International Electrotechnical Commission, Ec 62366-1:2015 — medical devices — part 1: Application of usability engineering to medical devices, <https://www.iso.org/standard/63179.html>, last accessed 14 January 2026 (2015).
- [73] Medical Research Council (MRC) Regulatory Support Centre and the Health Research Authority (HRA), Do I need NHS Ethics approval?, accessed 12 January 2025.
URL <https://www.hra-decisiontools.org.uk/ethics/>
- [74] Microsoft, Microsoft Teams (Version 2023.43.01.22 onwards) [Computer Software] (n.d.).
URL <https://www.microsoft.com/en-gb/microsoft-teams/group-chat-software>
- [75] K. E. Rudestam, R. R. Newton, *Surviving Your Dissertation*, 4th Edition, SAGE, Thousand Oaks (CA), 2015.
- [76] T. R. Lindlof, B. C. Taylor, *Qualitative Communication Research Methods*, 4th Edition, SAGE, London, 2017.
- [77] H. J. Rubin, I. S. Rubin, *Qualitative Interviewing: The Art of Hearing Data*, 2nd Edition, SAGE, London, 2005.
- [78] K. Kuutti, L. J. Bannon, The turn to practice in HCI: towards a research agenda, in: *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, ACM, Toronto Ontario Canada, 2014, pp. 3543–3552. doi:10.1145/2556288.2557111.

- [79] C. C. Voeten, Buildmer: Stepwise Elimination and Term Reordering for Mixed-Effects Regression (2025). doi:10.32614/CRAN.package.buildmer.
- [80] R Core Team, R: A Language and Environment for Statistical Computing, Vienna, Austria (2025).
- [81] D. Bates, M. Mächler, B. Bolker, S. Walker, Fitting Linear Mixed-Effects Models Using lme4, *Journal of Statistical Software* 67 (2015) 1–48. doi:10.18637/jss.v067.i01.
- [82] J. Peirce, J. R. Gray, S. Simpson, M. MacAskill, R. Höchenberger, H. Sogo, E. Kastman, J. K. Lindeløv, PsychoPy2: Experiments in behavior made easy, *Behavior Research Methods* 51 (1) (2019) 195–203. doi:10.3758/s13428-018-01193-y.
- [83] Prolific.co, Prolific (2024).
- [84] R. V. Lenth, Emmeans: Estimated Marginal Means, Aka Least-Squares Means (2025). doi:10.32614/CRAN.package.emmeans.
- [85] H.-C. Nothdurft, The role of features in preattentive vision: Comparison of orientation, motion and color cues, *Vision Research* 33 (14) (1993) 1937–1958. doi:10.1016/0042-6989(93)90020-W.
- [86] J. M. McCabe, E. J. Armstrong, I. Ku, A. Kulkarni, K. S. Hoffmayer, P. D. Bhave, S. W. Waldo, P. Hsue, J. C. Stein, G. M. Marcus, S. Kinlay, P. Ganz, Physician Accuracy in Interpreting Potential ST-Segment Elevation Myocardial Infarction Electrocardiograms, *Journal of the American Heart Association* 2 (5) (2013) e000268. doi:10.1161/JAHA.113.000268.
- [87] G. Veronese, F. Germini, S. Ingrassia, O. Cutuli, V. Donati, L. Bonacchini, M. Marcucci, A. Fabbri, The Italian Society of Emergency Medicine (SIMEU), Emergency physician accuracy in interpreting electrocardiograms with potential ST-segment elevation myocardial infarction: Is it enough?, *Acute Cardiac Care* 18 (1) (2016) 7–10. doi:10.1080/17482941.2016.1234058.

- [88] M. Tory, T. Moller, Human factors in visualization research, *IEEE Transactions on Visualization and Computer Graphics* 10 (1) (2004) 72–84. doi:10.1109/TVCG.2004.1260759.
- [89] A. J. Pugh, C. D. Wickens, N. Herdener, B. A. Clegg, C. A. P. Smith, Effect of visualization training on uncertain spatial trajectory predictions, *Human Factors* 60 (3) (2018) 324–339, pMID: 29498888. arXiv:<https://doi.org/10.1177/0018720818758770>, doi:10.1177/0018720818758770.
URL <https://doi.org/10.1177/0018720818758770>
- [90] A. Goldberger, L. Amaral, G. Glass, J. Hausdorff, P. C. Ivanov, R. Mark, J. E. Mietus, G. B. Moody, C. K. Peng, H. E. Stanley, PTB diagnostic ECG database v1.0.a (2004). doi:10.13026/C28C71.
- [91] C. L. Sattar Y, Electrocardiogram, in: StatPearls, StatPearls Publishing, Treasure Island (FL), 2023.
URL <https://www.ncbi.nlm.nih.gov/books/NBK549803/>
- [92] R. Avram, G. H. Tison, K. Aschbacher, P. Kuhar, E. Vittinghoff, M. Butzner, R. Runge, N. Wu, M. J. Pletcher, G. M. Marcus, J. Olgin, Real-world heart rate norms in the health eheart study, *NPJ Digit. Med.* 2 (1) (2019) 58. doi:<https://doi.org/10.1038/s41746-019-0134-9>.
- [93] P. Virtanen, R. Gommers, T. E. Oliphant, M. Haberland, T. Reddy, D. Cournapeau, E. Burovski, P. Peterson, W. Weckesser, J. Bright, S. J. van der Walt, M. Brett, J. Wilson, K. J. Millman, N. Mayorov, A. R. J. Nelson, E. Jones, R. Kern, E. Larson, C. J. Carey, Í. Polat, Y. Feng, E. W. Moore, J. VanderPlas, D. Laxalde, J. Perktold, R. Cimrman, I. Henriksen, E. A. Quintero, C. R. Harris, A. M. Archibald, A. H. Ribeiro, F. Pedregosa, P. van Mulbregt, SciPy 1.0 Contributors, SciPy 1.0: Fundamental algorithms for scientific computing in python, *Nature Methods* 17 (2020) 261–272. doi:10.1038/s41592-019-0686-2.
- [94] D. Makowski, T. Pham, Z. J. Lau, J. C. Brammer, F. Lespinasse, H. Pham, C. Schölzel, S. H. A. Chen, NeuroKit2: A Python toolbox for neurophysiological signal processing, *Behavior Research Methods* 53 (4) (2021) 1689–1696. doi:10.3758/s13428-020-01516-y.

- [95] Y. Luo, R. H. Hargraves, A. Belle, O. Bai, X. Qi, K. R. Ward, M. P. Pfaffenberger, K. Najarian, A Hierarchical Method for Removal of Baseline Drift from Biomedical Signals: Application in ECG Analysis, *The Scientific World Journal* 2013 (1) (2013) 896056. doi:<https://doi.org/10.1155/2013/896056>.
- [96] J. Zhang, Y. Guo, X. Dong, T. Wang, J. Wang, X. Ma, H. Wang, Opportunities and challenges of noise interference suppression algorithms for dynamic ECG signals in wearable devices: A review, *Measurement* 250 (2025) 117067. doi:<https://doi.org/10.1016/j.measurement.2025.117067>.
- [97] G. Lenis, N. Pilia, A. Loewe, W. H. W. Schulze, O. Dössel, Comparison of Baseline Wander Removal Techniques considering the Preservation of ST Changes in the Ischemic ECG: A Simulation Study, *Computational and Mathematical Methods in Medicine* 2017 (1) (2017) 9295029. doi:<https://doi.org/10.1155/2017/9295029>.
- [98] S. Butterworth, On the Theory of Filter Amplifiers, *Experimental Wireless & the Wireless Engineer* 7 (1930) 536–541.
- [99] R. Kher, Signal Processing Techniques for Removing Noise from ECG Signals, *Journal of Biomedical Engineering and Research* 3 (1) (2019) 1–9.
- [100] P. Kligfield, L. S. Gettes, J. J. Bailey, R. Childers, B. J. Deal, E. W. Hancock, G. van Herpen, J. A. Kors, P. Macfarlane, D. M. Mirvis, O. Pahlm, P. Rautaharju, G. S. Wagner, Recommendations for the standardization and interpretation of the electrocardiogram, *Circulation* 115 (10) (2007) 1306–1324. doi:<https://doi.org/10.1161/CIRCULATIONAHA.106.180200>.
- [101] S. Chatterjee, R. S. Thakur, R. N. Yadav, L. Gupta, D. K. Raghuvanshi, Review of noise removal techniques in ECG signals, *IET Signal Processing* 14 (9) (2020) 569–590. doi:<https://doi.org/10.1049/iet-spr.2020.0104>.
- [102] A. V. Rosen, S. Koppikar, C. Shaw, A. Baranchuk, Common ECG Lead Placement Errors. Part I: Limb lead Reversals, *International Journal*

- of Medical Students 2 (3) (2014) 92–98. doi:<https://doi.org/10.5195/ijms.2014.95>.
- [103] A. V. Rosen, S. Koppikar, C. Shaw, A. Baranchuk, Common ECG Lead Placement Errors. Part II: Precordial Misplacements, International Journal of Medical Students 2 (3) (2014) 99–103. doi:<https://doi.org/10.5195/ijms.2014.96>.
 - [104] D. L. Glancy, D. L. P. Jr., Inverted P Waves, Qrs Complexes, and T Waves in Lead I in a 64-Year-Old Woman, Baylor University Medical Center Proceedings 27 (2) (2014) 124–124. doi:<https://doi.org/10.1080/08998280.2014.11929082>.
 - [105] E. A. Ashley, J. Niebauer, Cardiology Explained, Remedica, 2004, Ch. 3: Conquering the ECG, pp. 15–33.
URL <https://www.ncbi.nlm.nih.gov/books/NBK2214/>
 - [106] L. Xie, Z. Li, Y. Zhou, Y. He, J. Zhu, Computational Diagnostic Techniques for Electrocardiogram Signal Analysis, Sensors 20 (21) (2020). doi:<https://doi.org/10.3390/s20216318>.
 - [107] D. G. Strauss, D. Schocken, Marriott’s practical electrocardiography, thirteenth edition. Edition, Wolters Kluwer, Philadelphia, PA, 2021.
 - [108] J. Martinez, R. Almeida, S. Olmos, A. Rocha, P. Laguna, A wavelet-based ecg delineator: evaluation on standard databases, IEEE Transactions on Biomedical Engineering 51 (4) (2004) 570–581. doi:10.1109/TBME.2003.821031.
 - [109] British Medical Association, Doctors’ titles explained, <https://www.bma.org.uk/advice-and-support/international-doctors/life-and-work-in-the-uk/toolkit-for-doctors-new-to-the-uk/doctors-titles-explained>, accessed 14 January 2026 (nd.).